

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: MLA0303

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO1: *Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems.(BL2, BL3)*

Assessment Tool Weightage: 30%

Dr Dumpala Prasanth

QUESTIONS: 10 Total Marks:50 Annexure A

Constraint - Based Problem-Solving Duration: 2Hrs

1. Develop a Python-based Reinforcement Learning model using the OpenAI Gym environment to solve a Grid World navigation problem where the agent must reach the goal while avoiding obstacles. Explain how the state space, action space, reward function, and constraints are defined.
(10 Marks)
2. Implement an ϵ -greedy Multi-Armed Bandit algorithm in Python to maximize rewards under a limited budget of 500 trials. Compare the effect of different ϵ values (0.1, 0.3, and 0.5) on exploration, exploitation, and constraint satisfaction.
(10 Marks)
3. Design a Markov Decision Process (MDP) for an autonomous robot that must reach a destination while minimizing energy consumption. Define the states, actions, transition probabilities, rewards, and energy constraints. **(10 Marks)**
4. Using Python and TensorFlow/Keras, implement a value iteration algorithm to determine the optimal policy for a constrained grid environment where certain states are restricted. Explain how Bellman equations are used to obtain the optimal policy.
(10 Marks)
5. A warehouse robot has a battery capacity of only 30 minutes. Design an RL framework that enables the robot to complete the maximum number of deliveries without exceeding the battery limit. Explain the constraints, policy, and reward design.
(10 Marks)
6. Implement a Reinforcement Learning agent that controls traffic signals at an intersection while ensuring that emergency vehicles receive immediate priority. Explain how exploration, exploitation, and policy learning are modified to satisfy the safety constraint.
(10 Marks)
7. Develop a Python program using Keras/TensorFlow to simulate a reinforcement learning agent for packet routing in a wireless network with limited bandwidth.

Explain how bandwidth constraints influence the reward function and policy optimization.

(10 Marks)

8. Analyse an experimental comparison between random action selection and ϵ greedy action selection in a Multi-Armed Bandit problem under a fixed time constraint. Record cumulative rewards and explain which strategy provides better performance.

(10 Marks)

9. Design a Reinforcement Learning framework for a delivery drone that must maximize successful deliveries while avoiding no-fly zones and maintaining battery limits. Explain the MDP formulation, Bellman equation, and policy learning process.

(10 Marks)

10. Implement a Reinforcement Learning solution using Python for a smart energy management system where electricity consumption must remain below a specified threshold. Explain how reward shaping, policy learning, and feasibility constraints improve decision-making.

(10 Marks)

Code of Conduct:

I Ch. Venkata Lakshmi Hymavathi (192425397) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Ch. Hymavathi

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints

Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

- ① Develop a python-based Reinforcement learning model using OpenAI Gym to solve a Grid world navigation problem. Explain the state space action space, reward function, and constraints.

Introduction

Reinforcement learning (RL) is a type of machine learning where an agent learns by interacting with an environment. The agent receives rewards for good actions and penalties for bad actions. In this problem, the agent moves in a grid world and learns the shortest safe path to reach the goal while avoiding obstacles.

Working

- (i) Agent starts from the initial position
- (ii) Chooses an action
- (iii) Receives reward
- (iv) Updates Q-values
- (v) Repeats until the goal is reached

Advantages:

- * learns optimal path automatically
- * Avoids obstacles
- * Improves performance after every episode

② Implement an ϵ -Greedy Multi-Armed Bandit algorithm and compare $\epsilon = 0.1, 0.3, 0.5$.

Introduction

The Multi-Armed Bandit problem helps an agent decide whether to explore new actions or exploit the best-known action. The ϵ -Greedy algorithm balances exploration and exploitation.

Working principle:

- * with probability ϵ the agent explores randomly
- * with probability $1-\epsilon$, the agent chooses the best action.

Advantages:

- * simple to implement
- * prevents the agent from getting stuck.
- * Finds better actions over time.

Design a Markov Decision process (MDP) for an autonomous robot minimizing energy consumption.

Introduction:

A Markov Decision process (MDP) is a mathematical model used to solve sequential decision making problems. The robot chooses actions that minimize energy while reaching the destination.

Components of MDP:

- * States
- * Actions
- * Transition probability
- * Energy constraints
- * Reward Function

Working:

- (i) observe current state
- (ii) Select action
- (iii) Move to next state
- (iv) Receive reward
- (v) Continue until destination.

Advantages:

- * Reduces battery consumption
- * Finds energy-efficient routes
- * Makes intelligent decisions

Q1) Implement value Iteration for a constrained Grid environment. Explain Bellman Equation.

Introduction:

Value Iteration is a Reinforcement learning algorithm that calculates the best value for every state. It uses the Bellman Equation repeatedly until the values become stable.

Working:

- (i) Initialize all state values to zero
- (ii) Update values using Bellman Equation.
- (iii) Repeat until values stop changing
- (iv) select the action with the highest value

Advantages

- * Finds optimal policy

- * Handles restricted environments
- * Improves navigation efficiency.

⑤ A warehouse robot has a battery capacity of only 30 minutes. Design a RL framework to maximize deliveries without exceeding the battery limit.

Introduction:

A warehouse robot must complete as many deliveries as possible before its battery runs out. Reinforcement learning helps robot learn the most efficient delivery strategy.

Working:

- (i) observe battery level
- (ii) select the next delivery
- (iii) Calculate shortest route
- (iv) Deliver package
- (v) Return for charging when needed.

Advantages

- * Increases delivery efficiency
- * Saves battery power.
- * Reduces travel time
- * Improves warehouse productivity.

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